J. Electroanal. Chem., 310 (1991) 335–359
Elsevier Sequoia S.A., Lausanne

The electrochemical desorption of n-alkanethiol monolayers from polycrystalline Au and Ag electrodes

Cindra A. Widrig, Chinkap Chung and Marc D. Porter *

Ames Laboratory–U.S. Department of Energy and Department of Chemistry, Iowa State University, Ames, IA 50011 (USA)

(Received 8 September 1990; in revised form 9 January 1991)

Abstract

Monolayers formed at Au and Ag surfaces by the spontaneous adsorption of n-alkanethiols were characterized voltammetrically to examine the chemistry of the bound thiol head group. Electrode reactions that correspond to the oxidative- and reductive-desorption of the adsorbed n-alkanethiol monolayer are reported for the first time. Analysis of the data indicates that upon adsorption at both Au and Ag surfaces the hydrogen of the thiol group is lost and the sulfur atom is oxidized by one electron. Based on the charge required for the reductive-desorption of the monolayer, the surface coverage of the oxidized n-alkanethiol species is 9.3×10^{-10} mol/cm^2 and 7.0×10^{-10} mol/cm^2 on Au and Ag, respectively. The value of the surface coverage at Au is slightly greater than that expected for a closest-packed overlayer commensurate with a Au(111) substrate, whereas the value at Ag is somewhat less than that expected for layers commensurate at any of the low index Ag surface planes. The low apparent coverage observed for Ag electrodes is attributed to a portion of the monolayer being electroinactive at accessible applied voltages. As a probe of the mechanism for monolayer formation, films that were deposited on Ag and Au surfaces from solutions containing sodium n-octadecanethiolate were also characterized. Examination of the resulting layers with infrared reflection spectroscopy and optical ellipsometry indicates that only at Ag substrates were the films deposited from the thiolate solution complete and structurally similar to monolayers formed from n-octadecanethiol solutions. Based on this observation, it is postulated that the thiol hydrogen of the n-alkanethiol molecule participates in the reduction reaction that is concomitant to the thiol oxidation during adsorption on Au, whereas adsorption on Ag proceeds through the reduction of the Ag(I) surface species of the native oxide. A model for the electric double layer at the monolayer-coated-electrode/solution interface is also suggested based on the observed chain-length dependence of both the capacitance and the potential for the reductive desorption.

* To whom correspondence should be addressed.

0022-0728/91/$03.50 © 1991 – Elsevier Sequoia S.A.

(I) INTRODUCTION

The chemical derivatization of metal surfaces is often considered as a means to control the interfacial reactivity of a metal in relation to processes such as adhesion, lubrication, corrosion, electrocatalysis, or electroanalysis [1]. Understanding the factors that govern the formation of stable derivatizing layers is therefore of both technological and fundamental importance. To this end, monomolecular films formed from surfactant molecules possessing a head group that binds to a particular metal and a tail group that possesses a specific chemical functionality have been examined as model interfacial structures. Factors such as the adsorbate–substrate registry, the head-group binding chemistry, the role of intermolecular interactions between neighboring adsorbate molecules, and the energetics of the adsorbate/air or -/liquid interface have been found relevant to the formation of stable surface films [2–6].

Organosulfur surfactants are known to bind strongly at a variety of metals such as iron, copper, gold, and silver, and monolayer films of n-alkanethiols on gold have been extensively studied in regard to their structure, electronic properties, and permeability to ion transport [7–11]. The chemistry describing the thiol–metal interaction, however, is not thoroughly understood. It has been shown by surface enhanced Raman spectroscopy that organothiol molecules upon adsorption at silver lose the hydrogen from the thiol group and that a S–Ag bond is formed [12–14]. The loss of the hydrogen has rationally been assumed to occur also with thiol adsorption at gold [10,15], but there is little supportive experimental evidence to this effect.

In this report, the electrochemical characteristics of n-alkanethiol monolayers adsorbed at Au and Ag films are described and discussed in relation to the head-group chemistry. The experiments are similar to those in which the redox properties of sulfide adsorbed at Au and Ag were examined [16–18]. At Au electrodes, cyclic voltammetry was used to examine the oxidation and reduction of the surface n-alkanethiol species and to assess the oxidation state of the sulfur atom in the thiol head group subsequent to adsorption. The observed ratio of the charges for the electro-oxidation and -reduction of the bound n-alkanethiol, together with the dependence of the potential for oxidation on the pH of the supporting electrolyte solution, indicate the loss of hydrogen from the thiol and the oxidation of the sulfur upon adsorption. We have also examined aqueous solutions that were used for the adsorption of n-propanethiol onto high surface area gold samples for products generated as a result of monolayer formation. The presence of hydrogen peroxide, formed from the reduction of dissolved molecular oxygen with evolved hydrogen, supports the postulate that the sulfur head group is oxidized upon adsorption. Furthermore, the inability to form complete monolayers at Au substrates from solutions that contain sodium n-octadecanethiolate indicates that the hydrogen of the thiol is involved in the reduction reaction.

The voltammetric experiments also provide a means by which the surface coverage of organothiols adsorbed at Au may be determined. For all n-alkanethiol

monolayers examined in this study ($C_nH_{(2n+1)}SH$ for $n = 3$–10, 12, 16, and 18), the surface coverage was found to be 9.3 (± 0.6) $\times 10^{-10}$ mol/cm^2. Since the actual surface area for our Au electrodes is 1.2 times the geometric surface area, this coverage is consistent with structures proposed from the results of helium diffraction [19], transmission electron diffraction [20], infrared reflection [6], and scanning tunneling microscopy experiments [21] in which monolayers from various long-chain n-alkanethiols were found to form as a $(\sqrt{3} \times \sqrt{3})R30°$ overlayer on Au surfaces of predominantly (111) character. Reductive desorption of n-alkanethiol films from Ag is also possible, though only n-propane-, n-butane-, n-pentane-, and n-hexanethiol monolayers are reduced at an accessible potential. The quantity of electroactive thiol adsorbed on Ag for each of these compounds is 7.0 (± 1.5) $\times 10^{-10}$ mol/cm^2. Though we do not fully understand the difference in the surface coverages measured on Au in comparison to Ag, we believe it results from a portion of the monolayer being electroinactive at accessible applied voltages.

Finally, it is shown that the behavior of the electrical double layer at the monolayer-coated-electrode/electrolyte interface can be modelled as that of two capacitors that are connected in series. This model is based on the chain-length dependence of both the capacitance of and the potential for the reductive desorption of the n-alkanethiol monolayers. The capacitance of the first capacitor is determined by the dielectric constant and thickness of the region of the monolayer occupied by the thiol head-group and that of the second capacitor by those properties of the hydrocarbon tail-group region. From these data, the conductivity of the hydrocarbon portion of the film is estimated roughly.

(II) EXPERIMENTAL

(II.A) Sample preparation

Substrate electrodes were prepared by the resistive evaporation of ~ 200 nm thick Au (99.9% purity) or Ag (99.9% purity) films onto 2″ (1″ = 2.54 cm) diameter polished p-type silicon (111) wafers (Virginia Semiconductors, Inc., Fredricksburg, VA), glass microscope slides (Fischer), or freshly cleaved 3″ × 1″ green mica sheets (Asheville-Schoomaker Mica Co., Newport News, VA). The evaporation rates, which were measured with a quartz thickness monitor, were 0.6 to 1.0 and 2.5 to 3.0 nm/s for Au and Ag, respectively. Approximately 20 nm of Cr was evaporated onto the Si wafers as an adhesive layer prior to Au deposition. The pressure in the cryopumped E360A Edwards Coating System during evaporation was $< 9 \times 10^{-5}$ Pa (7×10^{-7} Torr). After ~ 20 min cooling time, the evaporator was back-filled with dry N_2, and the substrates removed. The Au on mica films were subsequently annealed in the laboratory ambient at 300 °C for 3 h. Prior to derivatization, the Au surfaces were cleaned in freshly prepared and cooled chromic acid, rinsed thoroughly with deionized water, and dried in an air stream. The Ag electrodes were derivatized immediately upon their removal from the vacuum chamber.

Alkanethiol monolayers were deposited by their spontaneous adsorption from

ethanolic solutions [6]. Briefly, the substrates were immersed in approximately 1 mM solutions of the n-alkanethiol for 2 to 24 h, emersed, rinsed with ethanol, and allowed to dry. Liquid n-alkanethiols ($C_nH_{(2n+1)}SH$ with $n = 3$–10, 12, 16) were acquired from two sources (Aldrich, $n = 3$, 5, 7, 8, 9; Eastman Kodak, $n = 4$, 6, 10, 12, 16) and were purified by passage through an activity 1, neutral α-alumina column just prior to use. The n-octadecanethiol was obtained from Aldrich and recrystallized twice from methanol prior to use. All other chemicals were reagent grade unless otherwise specified.

(II.B) Electrochemical measurements

The Ag and Au electrodes were mounted in a conventional three-electrode cell with an exposed surface area of 0.67 cm^2, which was defined by the diameter of a circular opening cut into an inert gasketing material. Solutions were degassed with water-saturated Ar (Air Products) or N_2 (Air Products) for 15 min. All potential measurements are reported with respect to a Ag/AgCl/sat'd KCl electrode. Electrolyte solutions were prepared from conc. phosphoric acid and/or analytical grade K_2HPO_4 (Baker), semiconductor grade KOH (Aldrich), reagent grade NaOH, purified LiOH, and/or reagent grade KCl (all Fischer).

(II.C) Determination of H_2O_2

To determine the presence of H_2O_2 in the adsorption solutions subsequent to the formation of monolayers on high-surface-area gold samples, a spectrophotometric assay was used. The gold sample was a powder composed of 2 to 5 μm spheres (Alfa) with an approximate surface area of ~ 900 cm^2/g (assuming the average particle diameter is 3.5 μm). Prior to monolayer formation, 0.45 g of the Au powder was suspended in hot concentrated nitric acid for 30 min and then rinsed with deionized water until the pH of the wash was greater than 6. The powder was then suspended in 3 ml of O_2 saturated water and 8×10^{-7} mol of n-propanethiol was added. The mixture was stirred for 30 to 60 min after which time the supernatant was removed and purged with Ar for 20 min to remove dissolved oxygen and residual n-propanethiol. Concentrated HCl (0.5 ml) was added to acidify the solution and the mixture was purged for an additional 10 min. A quantity of potassium iodide sufficient to saturate the solution was added to reduce any H_2O_2 present. The initial concentration of H_2O_2 is reflected in the final concentration of triiodide, which is determined from the solution absorbance at 352 nm. Using this method, a detection limit for H_2O_2 of 0.1 μM is estimated, whereas a final concentration of H_2O_2 in the adsorption solution would be approximately 60 μM based on the probable surface coverage and chemisorptive process in forming the layers (see below).

(II.D) Monolayer preparation from solutions containing sodium n-octadecanethiolate

Sodium n-octadecanethiolate was prepared by the addition of a slight excess of sodium ethoxide to an ethanol solution containing n-octadecanethiol. The solution

was stirred at room temperature for several hours and the solvent was evaporated. The solid residue was rinsed several times with n-hexane to remove residual n-octadecanethiol, recrystallized from methanol, and dried in vacuum. Substrates of Au/mica and Ag/mica were exposed to solutions of n-octadecanethiol and sodium n-octadecanethiolate in purified THF [22]. After removal from solution, the samples were washed with hexadecane and examined by infrared reflection spectroscopy and optical ellipsometry. All reactions were carried out in a glove bag in an atmosphere of dry nitrogen.

(II.E) Instrumentation

Cyclic voltammetry was performed with a CV-27 potentiostat (Bioanalytical Systems, Inc., West Lafayette, IN). The surface of the evaporated metal films were examined in air with a NanoScope II (Digital Instruments, Inc., Santa Barbara, CA) scanning tunneling microscope. Contact angles between the derivatized surface and hexadecane or deionized water were measured in air with a Rame-Hart Model 100-00 115 goniometer. Infrared spectra were collected using p-polarized light at grazing incidence with a Nicolet 740 FT-IR spectrometer (Nicolet Analytical Instruments, Madison, WI). Reference spectra were collected from uncoated Au/mica electrodes that were cleaned in fresh chromic acid and rinsed in deionized water immediately prior to use. Monolayer thicknesses that are discussed were determined by optical ellipsometry using a Gaertner Model L-116B Auto Gain ellipsometer at a wavelength of 632.8 nm (He–Ne laser). To calculate the thicknesses, a refractive index of 1.45 for the hydrocarbon film was assumed. The surface composition of the underivatized substrates was examined by Auger electron spectroscopy (AES) in a PHI Auger Multiprobe 600 system.

(III) RESULTS AND DISCUSSION

(III.A) Substrate characterization: roughness and crystallinity

The results described in the following sections were obtained for samples prepared by the spontaneous adsorption of monolayers of n-alkanethiols onto thin polycrystalline films of Ag or Au. Because the nature of the substrate surface appears to effect the electrochemical behavior of the monolayer films, we will first discuss the surface topography and crystallinity of the various substrates used. From underpotential deposition studies and a survey of the current literature examining vapor deposited films [23–26], it is apparent that the surfaces of the metal layers are, in most cases, chemically heterogeneous.

The majority of the work that will be discussed used as the substrate electrode Au films that were annealed at 300°C in the laboratory ambient subsequent to their vapor deposition on mica (Au/mica electrodes). A less detailed examination of monolayers on unannealed Au films at silicon wafers and glass slides (Au/Si and Au/glass electrodes) as well as unannealed Ag films at mica, silicon, and glass

(Ag/mica, Ag/Si, and Ag/glass electrodes) was also undertaken. Scanning tunneling microscopy (STM) was used to examine the topography of the various substrates. Representative images of 450 × 450 nm sections of the substrate surfaces are shown in Figs. 1A through 1D. The conditions at which the images were obtained are given in the figure captions. It can be seen that the Au/mica surface (Figure 1A) is composed of small (~ 100 nm) atomically terraced crystallites separated by grain boundaries. A more detailed description of annealed Au/mica surfaces has previously been presented [27]. Both Ag/mica, Au/Si, and Ag/Si electrodes (Figures 1B, 1C, and 1D, respectively) are made up of crystallites that are both smaller (~ 25 nm) and rougher than those on the Au/mica surface. The surfaces of Au/glass and Ag/glass electrodes resemble topographically those of Au/Si and Ag/Si, respectively.

The predominant atomic surface structure for the Au/mica films is that of the (111) indexed plane. This was determined by atomic resolution STM [21,28] and by

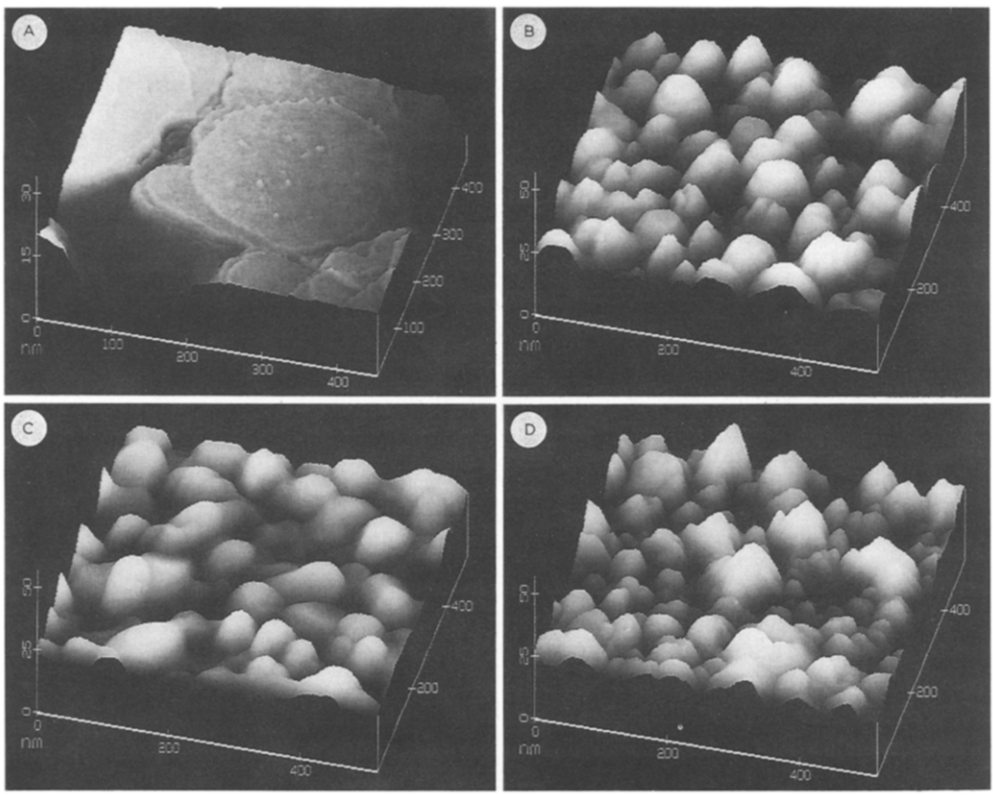


Fig. 1. Representative images obtained by Scanning Tunneling Microscopy of 450 nm × 450 nm sections of (A) Au/mica, (B) Ag/mica, (C) Au/Si, and (D) Ag/Si surfaces. The microscope was operated in a constant current mode. The bias voltage was maintained at 80 mV (tip relative to sample) and the tunneling current at 3 nA. The data were filtered through a lowpass filter to remove high frequency noise. Tips were fabricated from mechanically cut tungsten wire.

comparing the potential and shape of the voltammetric waves for the underpotential deposition and stripping (UPD) of Pb from the underivatized Au surface to those for the UPD of Pb at single crystal Au surfaces [23] *. The surfaces of both Au/Si and Au/glass electrodes showed mostly Au(111) character but there was also evidence of Au(110) sites at these surfaces [23] *. Domains of Au(111) and Au(110) on Au films vapor deposited on glass supports were also found in electron diffraction studies of those surfaces [26]. For Ag surfaces, UPD experiments with Tl indicated that the Ag deposits on all three substrate materials were composed of Ag(100) and/or Ag(110) domains **. Previously reported electron diffraction studies of vapor deposited Ag films indicated surfaces of mixed crystallinity [23] and impedance measurements in the presence of chloride using evaporated Ag surfaces pointed to predominantly (110) character [25]. Other reports showed that the surface crystallinity of evaporated Ag films is very sensitive to the deposition conditions used [26]. Auger spectroscopy indicated that the "bare" Au and Ag substrates used in our studies were covered with monolayer quantities of chlorine, iodine, and/or oxygen. This contamination is presumed to be due to the adventitious adsorption of ambient impurities except for the case of oxygen on Ag, which results from the formation of a surface oxide on exposure of the freshly evaporated sample to the laboratory atmosphere.

(III.B) Monolayer characterization: structure and interfacial properties

Monolayers of n-alkanethiols at Ag/Si and Au/Si surfaces have been examined extensively in regard to their structure and physical properties [7–15,31]. To verify that the n-alkanethiol monolayers on Au/mica and Ag/mica are structurally comparable to those at Au/Si and Ag/Si, the monolayers formed were characterized with a variety of surface techniques including infrared reflection absorption spectroscopy (IRRAS), contact angle measurements, and electrochemical capacitance measurements. The C–H stretching region of infrared spectra of n-octadecanethiol monolayers at Au/mica and Ag/mica substrates are shown in Figs. 2A and 2B. The labeled peaks [32] are for the asymmetric and symmetric methylene (ν_{a,CH_2} and ν_{s,CH_2}) and methyl stretches (ν_{a,CH_3}, ν_{s,CH_2,FR_1}, and ν_{s,CH_2,FR_1}, where the latter two modes are from Fermi resonance between symmetric and umbrella stretches of the methyl group). Both the absolute and relative intensities of these peaks indicate that the surface coverage and structure within the hydrocarbon portion of the film at Au/mica and Ag/mica are comparable to those for n-octadecanethiol layers at Au/Si [7] and Ag/Si [31] surfaces, respectively. Specifically, the data indicate that on both surfaces the average hydrocarbon chain is fully

* Voltammograms for the UPD of Pb were recorded at the vapor Au deposited electrodes and compared to CV's for UPD of Pb at single crystal Au surfaces as given in ref. 29.

** Voltammograms for the UPD of Tl were recorded at the vapor deposited Ag electrodes and compared to CV's for UPD of Tl on single crystal Ag surfaces given in ref. 30.

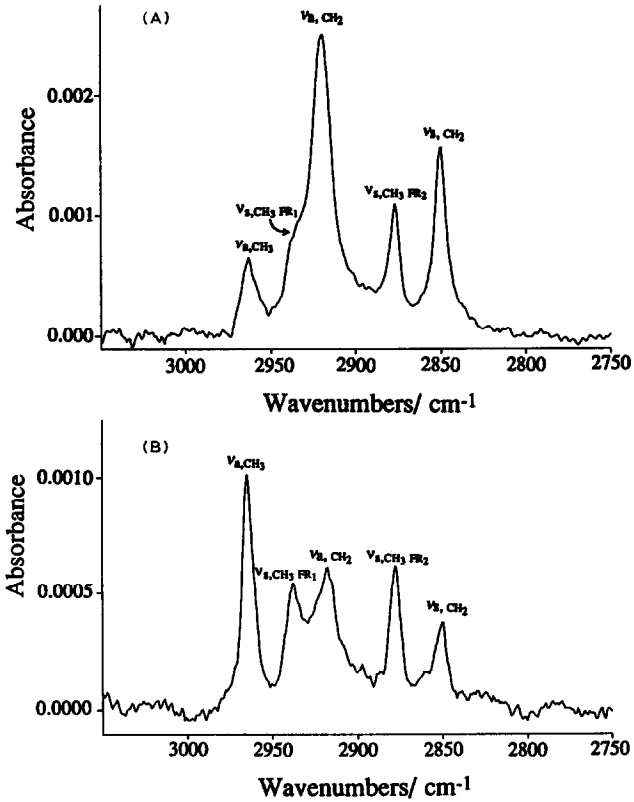


Fig. 2. Infrared reflection absorbance spectra for n-octadecanethiol monolayers adsorbed at (A) Au/mica and (B) Ag/mica surfaces.

extended in a predominantly trans zig-zag conformation and tilted ~ 28° from the surface normal on Au/mica and ~ 10° on Ag/mica. The differences in the chain tilts can be qualitatively understood from consideration of the infrared surface selection rule which results in the preferential excitation of vibrational modes with transition dipoles that are normal to the metal surfaces [33]. For the methylene vibrations the transition dipoles are perpendicular to the hydrocarbon chain. Therefore, the greater the angle of tilt of the hydrocarbon chain away from surface normal, the larger the perpendicular component of the methylene transition dipoles, and the greater the absorbance for those modes. As such, the relatively lower absorbances of the methylene modes at Ag as compared to Au indicate that the alkyl chains at Ag are aligned more along the surface normal and are more densely packed than those at Au.

As noted, both the contact angles and the electrochemical capacitances of the layers at our Au-coated mica samples were also measured. The advancing contact angles of water and hexadecane at the hydrocarbon/air interface of one n-oc-

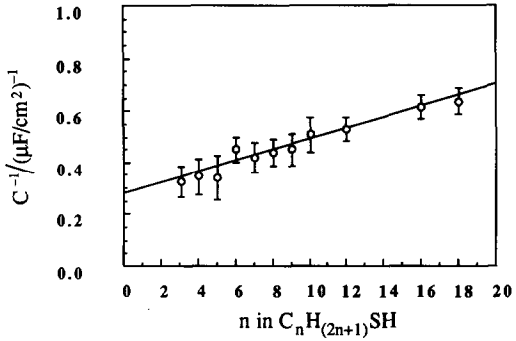


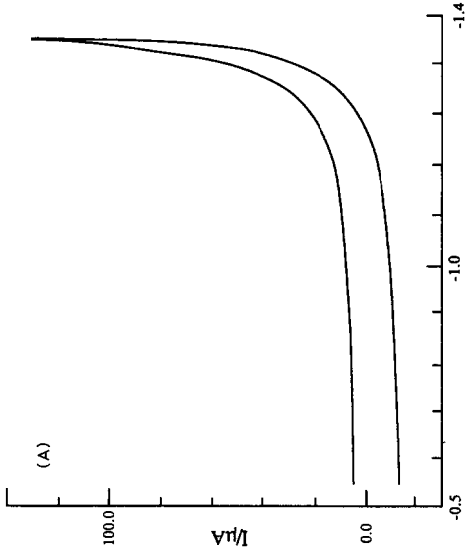
Fig. 3. Reciprocal capacitance vs. n for $C_{2n}H_{(2n+1)}SH$ monolayer coated Au/mica surfaces. The capacitances were determined from the charging currents at 0.0 V for cyclic voltammograms in which the electrode was cycled between +0.2 and −0.4 in at 100 min/s in 0.5 M KOH. In this potential range, the charging current was independent of the applied potential. Each data point represents the average of reciprocal capacitance of approximately 10 electrodes. The error bars are extended ±1 standard deviation.

tadecanethiol layer on both Au/mica and Ag/mica are 109° and 49°, respectively. These values are typical for a densely packed array of n-alkyl chains and agree with those previously reported for comparable structures [7,8,31]. In Fig. 3, the reciprocal capacitances of the monolayer coated Au/mica electrodes are plotted as a function of n, the number of carbons in the hydrocarbon chain. These data are calculated from the charging currents at 0.0 V of cyclic voltammograms recorded in 0.5 M KOH at 100 mV/s. The plot of $1/C$ vs. n is reasonably described as a straight line with a slope of 0.02 $(\mu F/cm^2)^{-1}/n$ and intercept of 0.28 $(\mu F/cm^2)^{-1}$. The linear dependence of $1/C$ on n qualitatively agrees with previously reported capacitance data at Au/Si [6]; interestingly, our values are slightly lower than those at Au/Si which suggests that our layers may be more permeable. However, taken together, the results of the infrared, contact angle, and capacitance experiments indicate comparable structures for monolayers on metal films supported at mica and Si.

(III.C) Head-group chemistry for n-alkanethiols adsorbed at Au/mica electrodes

(III.C.1) Reductive desorption of n-alkanethiol monolayers from Au/mica electrodes

Figure 4A shows a cyclic voltammogram in the potential range from −0.5 to −1.4 V recorded a bare Au/mica electrode in 0.5 KOH. Figures 4B and 4C show the first (solid curve) and second (dashed curve) potential scans with n-hexanethiol and n-octadecanethiol coated Au/mica electrodes recorded under conditions identical to those of Fig. 4A. In each case, the potential scans were initiated at 0.0 V vs. Ag/AgCl/sat. KCl and continued at −100 mV/s to the limit of solvent reduction. We attribute the large cathodic peak observed in the first potential cycle in Figs. 4B and 4C to the reduction of the adsorbed n-alkanethiol molecules. These peaks are positive of the potential for the onset of solvent reduction and the peak potential for



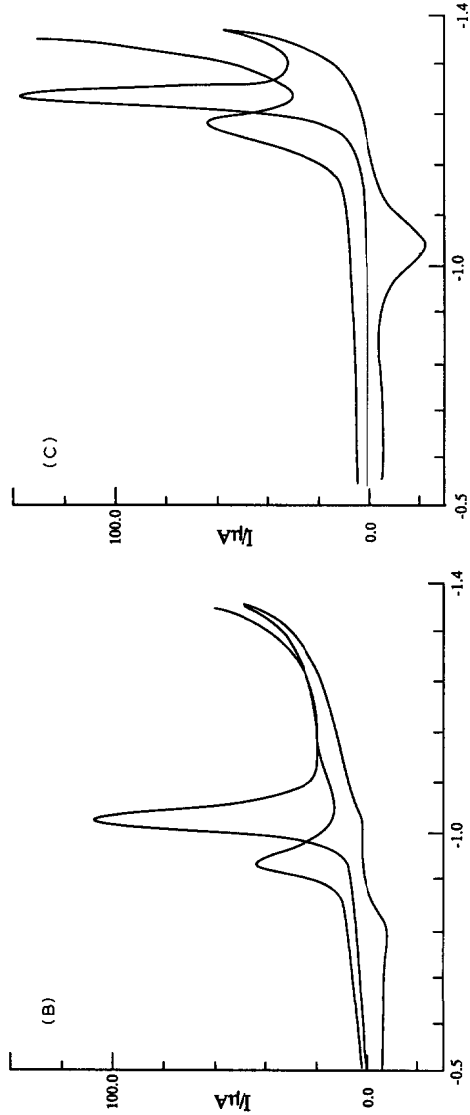


Fig. 4. Cyclic voltammograms initiated at 0.0 V for (A) bare Au/mica electrode and for the reductive desorption of (B) n-hexanethiol (C) n-octadecanethiol monolayers from Au/mica surfaces. The voltammograms were recorded at 100 mV/s in 0.5 M KOH. In (B) and (C), the solid and dashed curves represent the first and second voltammetric scans, respectively. The electrode area is 0.67 cm^2.

the reductive desorption of n-hexanethiol is positive of that for n-octadecanethiol. The charge associated with this reduction, σ_{rd}, was found to be 90 (± 7) $\mu C/cm^2$ and was independent of n for all n-alkanethiols tested ($n = 3$–10, 12, 16, 18). The reported value of σ_{rd} was determined by integration of the cathodic peak in the first voltammetric scan and is the average from at least 10 monolayer coated electrodes for each of the n-alkanethiols.

If we note that the actual surface area of the Au/mica electrode is 1.2 times its geometric surface area and if it is assumed that the surface coverage of the n-alkanethiol is 7.6×10^{-10} mol/cm^2, the charge found for the reductive desorption of the n-alkanethiol monolayer at Au is consistent with the electrode reaction

$$AuSR + 1\ e^- \rightarrow Au(0) + RS^- \qquad (I)$$

A coverage of 7.6×10^{-10} mol/cm^2 is suggested because it is the value corresponding to the $(\sqrt{3} \times \sqrt{3})R30°$ overlayer structure for long-chain n-alkanethiols adsorbed on Au(111) surfaces as determined by helium diffraction [19], electron diffraction [20], and scanning tunneling microscopy [21] studies. This coverage corresponds to a hexagonal array of close-packed n-alkanethiol adsorbate molecules in which the nearest neighbor spacing is ~ 0.50 nm and the hydrocarbon chain is fully extended and tilted at ~ 30° relative to the surface normal. A roughness factor of 1.2 for the Au/mica electrodes is reasonable, as can be seen by examination of STM images of the electrode surface like that shown in Fig. 1A *. Reaction (I) implies both the loss of the thiol hydrogen and the oxidation of the S to a formal oxidation state of (−1) upon adsorption of the n-alkanethiol at Au. This conclusion is supported by the results of experiments in which the adsorbed alkanethiol is oxidatively desorbed from the electrode surface, as described in the following section. The product of reaction (I) is written as the n-alkanethiolate because the pH of the electrolyte solution is greater than the pK_a of an n-alkanethiol (~ 9 [34]).

The voltammetry observed subsequent to initial reduction of the adsorbate is also consistent with reaction (I). In the positive scan of the first potential cycle, partial re-oxidation of the generated reduction product is apparent. The product of this oxidation may be either the original surface species or the corresponding dialkyl disulfide. In the second cycle, the re-oxidized material is reduced at potentials more positive than the potential for reduction of the original surface species. The quantity of material re-oxidized is greater for higher scan rates where there is less time for the thiolate product to diffuse away from the electrode surface after reduction. When comparing voltammograms recorded at a given scan rate, the quantity of material re-oxidized is greater for monolayers of long-chain n-alkanethiols because the reduced n-alkanethiolate is more likely to precipitate at the electrode surface. This

* A roughness factor between 1.1 and 1.2 is determined by dividing the "actual" surface area of the electrode by its geometric area. The actual surface area is estimated to equal the square of length of the contour of a cross-section of an STM image of the Au/mica surface.

can be seen by comparing the second sweeps of the voltammograms Figs. 4B and 4C.

(III.C.2) Oxidative desorption of n-alkanethiol monolayers from Au/mica electrodes

Further evidence of the oxidation of the n-alkanethiol upon its adsorption at Au is found in the voltammetry for the oxidative desorption of n-propane and n-butane thiol monolayers from Au/mica surfaces. Figure 5 is a cyclic voltammogram for the oxidative desorption of n-propanethiol from Au/mica electrodes in 0.5 M KOH. The solid curve represents the first potential scan and the dashed curve the second (and subsequent) scans. Voltammograms similar to the one shown were also obtained for n-butanethiol films, however the oxidation of longer chain thiols occurred at potentials positive of the solvent oxidation. Uncoated Au/mica electrodes produced voltammograms identical to that given by the dashed curve in Fig. 5. To obtain the voltammograms in Fig. 5, the derivatized electrode was polarized at -0.5 V and its potential cycled at 100 mV/s between -0.50 and 0.80 V. On the first cycle, a large anodic peak is apparent positive of the potential at which the surface layer of the gold is normally oxidized. This wave is due to the oxidation of the surface bound n-alkanethiol and partial oxidation of the outer layer of the Au surface. On the reverse scan, the Au oxide is reduced. Subtraction of the integrated charge for oxide reduction from the total anodic charge in the first positive scan gives the charge for oxidation of the surface n-propanethiol, σ_{od}. On examination of more than 20 n-propanethiol coated electrodes (and 5 n-butanethiol samples), σ_{od} was determined to be 280 (± 35) $\mu C/cm^2$ and was independent of the electrolyte pH.

For n-propanethiol monolayers, the dependence of the potential for the oxidative desorption on the pH of the supporting electrolyte solution was examined to assess the relative number of protons and electrons involved in the desorption reaction. The potential of the oxidative desorption of n-propanethiol is plotted as a function of pH in Fig. 6. These data are from voltammograms recorded at 100 mV/s in 0.5 M phosphate buffers. It can be seen that there are two regions in which the potential of the desorption is linear with pH; at low pH (< 4), the slope is -52 mV/pH unit and at high pH (> 7), the slope is -79 mV/pH unit. These slopes are consistent with the 3 electron oxidation of RS to RSO_2H at pH < 4 and RSO_2^- at pH > 7, as given in reactions (II) and (III):

$$AuSR + 2\ H_2O \rightarrow Au(0) + RSO_2H + 3\ e^- + 3\ H^+ \quad \text{(II)}$$

$$AuSR + 2\ H_2O \rightarrow Au(0) + RSO_2^- + 3\ e^- + 4\ H^+ \quad \text{(III)}$$

A comparison of the charges for oxidative and reductive desorption of the n-propanethiol ($\sigma_{od}/\sigma_{rd} = 3.1 \pm 0.5$) is also consistent with the proposed reactions (I)–(III). Taken together, the results of the oxidative-and reductive-desorption experiments indicate that upon adsorption the thiol group of the n-alkanethiol is deprotonated and its S atom oxidized by one electron. The adsorption of organothiols at Au is therefore analogous to the adsorption of sulfide from aqueous solutions, which has been shown to result in the formation of monomolecular layers of atomic sulfur [16].

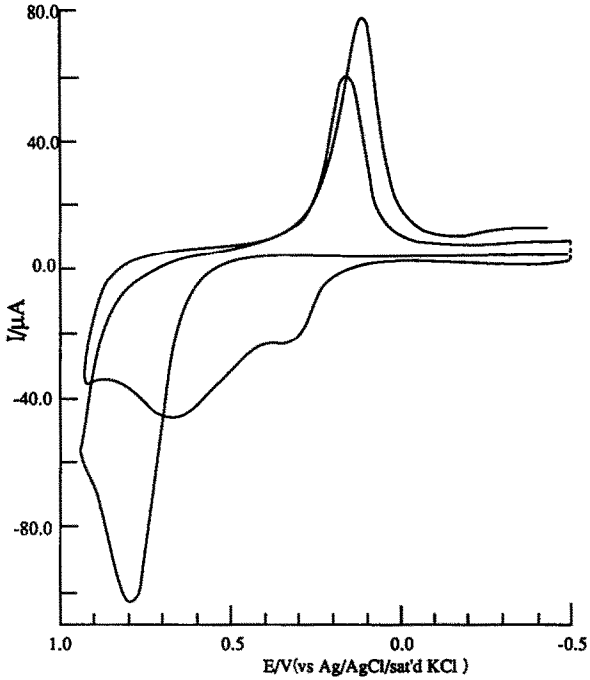


Fig. 5. A cyclic voltammogram for the oxidative desorption of an n-propanethiol monolayer from a Au/mica surface recorded at 100 mV/s in 0.5 M KOH. The solid and dashed curves represent the first and second voltammetric scans respectively. The electrode area is 0.61 cm^2.

(III.C.3) Determination of H_2O_2 in n-alkanethiol adsorption solutions

For the sulfur of the thiol head group to undergo oxidation upon adsorption of an n-alkanethiol molecule at gold, it is required that a simultaneous reduction occur

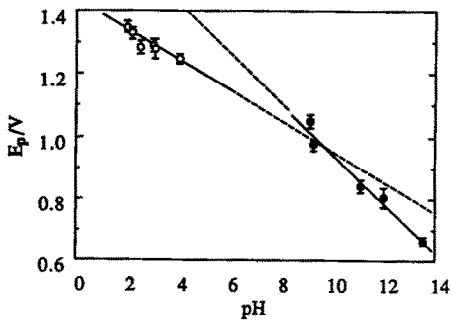


Fig. 6. The peak potential vs pH for the oxidative desorption of n-propanethiol from Au/mica surfaces. Data are from cyclic voltammograms recorded at 100 mV/s in constant ionic strength electrolytes made from H_3PO_4 and KOH. Each data point represents the average of the peak potential for desorption of more than 6 electrodes. The error bars shown indicate ± 1 standard deviation. The potentials are reported vs. Ag/AgCl/sat'd KCl.

in the adsorption solution. We therefore examined aqueous solutions of n-propanethiol that had been used for adsorption onto high surface area Au samples for possible reduction products. In oxygenated adsorption solutions, hydrogen peroxide was found to be present. In control experiments where (1) n-propanethiol was omitted from the adsorption solution and (2) Au samples previously derivatized with n-alkanethiols were exposed to n-propanethiol solutions, peroxide was not detected. This provides strong evidence that the observed peroxide results from the the adsorption of n-propanethiol at the high-surface-area Au substrate, and probably arises from the combination of H_2 and O_2.

The quantity of peroxide detected was at most 40% of that expected for the two electron reduction of 1 mol O_2 per 2 mol n-alkanethiol adsorbed. However, in view of the uncertainty of the actual surface area of the gold powder used, it is difficult to speculate on the significance of this discrepancy. Other complications, such as the consumption of H_2O_2 in the oxidation of solution n-alkanethiol to the dialkyl disulfide or the occurrence of competing reduction reactions, may affect the H_2O_2 concentration. In regard to the latter of these possibilities, we suspect that the reduction of H^+ to H_2 occurs during the adsorption of n-propanethiol from deaerated aqueous solutions. Monolayers formed from such solutions on Au/mica substrates exhibit voltammetry identical to the curves shown in Figs. 4B and 5, indicating oxidation of the sulfur and loss of the thiol hydrogen, but the pH of the adsorption solution during n-propanethiol deposition at the Au powder remained constant; if the adsorption of the n-alkanethiol resulted only in the loss of the thiol hydrogen as a proton, the pH of the reaction solution would fall to ~ 4 for the estimated surface area of the Au powder and the solution volume used in these experiments. These observations suggest the consumption of the thiol hydrogen as the oxidant upon adsorption from dearated reaction solutions. In summary, we propose that the adsorption of n-alkanethiols at Au/mica from aqueous solutions is described by:

$$2\ Au + 2RSH \rightarrow 2\ Au\text{–}SR + H_2 \qquad (IV)$$

and/or

$$2\ Au + 2\ RSH + O_2 \rightarrow 2\ Au\text{–}SR + H_2O_2 \qquad (V)$$

Reaction (IV) occurs in both the oxygenated and deoxygenated adsorption solutions, with the H_2O_2 formed in reaction (V) from the combination of H_2 from reaction (IV) and O_2.

(III.C.4) The formation of monolayers at Au/mica from solutions containing sodium n-octadecanethiolate

The conclusions from the above experiments pertain to adsorption of thiols from aqueous solutions and therefore are not straightforwardly applicable to the description of thiol adsorption from non-aqueous solutions, the reaction conditions most commonly employed. Although the voltammetry for n-alkanethiol layers adsorbed at Au/mica from ethanol solutions indicate oxidation of the thiol group, such

oxidation could occur upon exposure of the derivatized electrode to the aqueous electrolyte solution. The results of the following experiments in which sodium n-octadecanethiolate was used as the adsorbate precursor instead of n-octadecanethiol, however, argue that this is not the case and oxidation of the thiol group also occurs upon adsorption from organic solvents.

Au/mica electrodes were treated in deaerated THF solutions containing either n-octadecanethiol or sodium n-octadecanethiolate in a manner consistent with that for the formation of n-alkanethiol monolayers from ethanol solutions. The resulting films were examined by IRRAS and optical ellipsometry. The latter technique indicated that the monolayers formed from the n-octadecanethiol solution were of a reproducible and typical thickness, ~ 2.6 nm, but that films formed from the n-octadecanethiolate solution were invariably thinner, ~ 0.5 to 1.0 nm. Infrared reflection spectra also indicated that the structure of the resulting films from the n-octadecanethiol solutions were very similar to layers prepared from n-octadecanethiol in ethanol but that films formed from n-octadecanethiolate were less organized and of a lower coverage. Specifically, the absolute absorbances of all of the C–H stretching modes for the monolayer formed from the thiolate solution were less than those for the monolayer formed from the thiol. This indicates a lower surface coverage of the adsorbed n-alkanethiol when the monolayer is formed from the former solution. Furthermore, the relative absorbances of the methylene and methyl modes for monolayers formed from the thiolate indicated an isotropic dispersion of the C–H stretching transition dipoles, which points to a disordered chain structure. Together these observations indicate that the films formed from the thiolate species are of a lower coverage than those from n-octadecanethiol solutions. This result argues further that in dearated, aprotic adsorption solutions the thiol hydrogen is involved in the adsorption reaction in a manner consistent with reaction (V) and oxidation of the thiol does occur upon adsorption.

(III.D) A model for the double layer structure at n-alkanethiol coated Au/mica electrodes

The capacitive behavior of an n-alkanethiol coated electrode has been described as that of an ideal capacitor for which the reciprocal capacitance of unit area is:

$$C^{-1} = d/\epsilon\epsilon_0 \tag{1}$$

where d and ϵ correspond to the thickness and the dielectric constant of the n-alkanethiol monolayer, respectively, and ϵ_0 is the permitivity of free space [7]. According to this equation, C^{-1} should increase linearly with n, the number of carbons in the hydrocarbon tail of the n-alkanethiol, because:

$$d = (1.3 \times 10^{-8})n \cos\theta \tag{2}$$

The constant in eqn. (2) is the distance in cm between adjacent carbons in a fully extended methylene chain and θ is the tilt angle of the hydrocarbon chain with respect to the surface normal. As shown in Fig. 3, a linear dependence of C^{-1} on n

was observed. These data were determined from the charging currents at 0.0 V of cyclic voltammograms recorded with the derivatized electrode exposed to solutions of 0.5 M KOH. The slope and intercept of the least-square fit to the data are 0.02 $(\mu F/cm^2)^{-1}/n$ and 0.28 $(\mu F/cm^2)^{-1}$, respectively. Although eqn. (1) predicts the linear dependence, it does not provide a complete description of the capacitance of the interfacial region as the non-zero intercept of the best-fit line for the data in Fig. 3 is not predicted. The intercept corresponds physically to the reciprocal capacitance of a monolayer coated electrode for n equal to zero, i.e. the reciprocal capacitance of the surface-bound thiolate, coupled in a series arrangement with that of the film–solution interphase. A more adequate description for the electrical double layer at these electrodes is therefore three capacitors connected in series where the capacitance of the first is determined from d and ϵ of the thiol head-group, that of the second from d and ϵ of the hydrocarbon tail, and that of the third by the properties of the end group–solution interphase. The overall reciprocal capacitance, C_{Total}^{-1}, is given by [35]:

$$C_{Total}^{-1} = C_{MS}^{-1} + C_{HC}^{-1} + C_{DL}^{-1} \tag{3}$$

where C_{HC} and C_{MS} are the capacitances of the hydrocarbon and thiol portions of the film, and C_{DL} is the capacitance of solution-end-group interphase. From Figure 3, $C_{MS}^{-1} + C_{DL}^{-1}$ equals 0.28 $cm^2\ \mu F^{-1}$. If, however, the majority of the potential is dropped across the dielectric layer, as is indicated for these layers with long chain lengths by the independence of C_{Total} with respect to the composition and concentration of supporting electrolyte [7], then there will be a negligible space-charge at the film–solution interface. Such a situation resembles that described by Gouy–Chapman (GC) theory at the potential of zero charge (pzc), the capacitance of which is given for a 1 : 1 electrolyte at 25 °C as $C_{DL} = 228c^{-1/2}$, where c is the concentration of the electrolyte [36]. Based on GC theory, as well as on experimental data at the pzc [36] it is apparent that the contribution of C_{DL} to C_{Total} is small and eqn (3) becomes

$$C_{Total}^{-1} = C_{MS}^{-1} + C_{HC}^{-1} \tag{4}$$

Thus, C_{MS} is effectively equal to 3.5 $\mu F/cm^2$ and C_{HC} is equal to $50/n$ $\mu F/cm^2$.

The capacitances calculated above, according to eqn. (1), indicate that ϵ for the head-group region of the film $\cong 2$ (assuming that the thickness of the Au–S portion of the film is 4×10^{-8} cm) and ϵ for the hydrocarbon region is equal to 7. The latter ϵ is larger than that determined in previous electrochemical studies of n-alkanethiol films at Au [7] and studies of n-alkanoic acid monolayers in which the monolayer film was used as a tunnel junction between two Al plates [37]. In both of these investigations, ϵ was found to equal ~ 2.5. This difference may indicate that the monolayer films at Au/mica contain more defects than those in the cited studies. A simple electrical circuit model of the defective interfacial structure would therefore include a parallel capacitive component where the capacitance of the monolayer coated surface would be in parallel with that of a uncoated Au surface. At the present, we do not understand the source of these defects.

The potential change between the electrode surface and the outer boundary of the monolayer film is equal to the difference between the applied electrode potential and the solution potential. In consideration of eqn. (3), however, the potential gradient through the n-alkanethiol monolayer will not be uniform. For any applied potential, E_{app}, the potential drop across the head- and tail-group regions of the monolayer film, ΔE_{MS} and ΔE_{HC}, respectively, can be calculated according to [35]:

$$\Delta E_{MS} = (E_{sol} - E_{app})(C_{MS}^{-1}/C_{Total}^{-1}) \quad (5)$$

and

$$\Delta E_{HC} = E_{sol} - E_{app} - \Delta E_{MS} \quad (6)$$

The non-uniform potential gradient is manifested in the chain-length dependence of the peak potential for the reductive desorption of these monolayers as discussed below.

The peak potential, E_p, for the reductive desorption of an n-alkanethiol monolayer adsorbed at Au and Ag is plotted as a function of n in Fig. 7. The data shown are from voltammograms recorded at 100 mV/s in 0.5 M KOH supporting electrolyte. Discussion of the reductive desorption from Ag electrodes is deferred to Section III.F.1. The dependence of E_p on n for desorption from Au/mica is reasonably described by a straight line with a slope of -20 mV/n and an intercept of -0.83 V. We do not believe that the change in E_p reflects differences in the reversible reduction potential of the layers as n increases. Two experimental observations support our contention: E_p shifts to more positive values for both partially formed monolayer and bilayer structures. The first observation comes from experiments where the formation of the n-alkanethiol layers were halted at various emersion times before the formation of a densely packed structure. For all chain

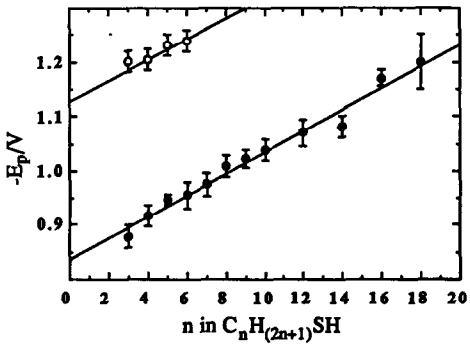


Fig. 7. The peak potential (E_p) vs. n for the reductive desorption of spontaneously adsorbed n-alkanethiol monolayers from Au/mica electrodes (●) and Ag/mica (○) electrodes. The value of n corresponds to the number of carbons in the alkanethiol adsorbate, $C_nH_{(2n+1)}SH$. The data are from cyclic voltammograms recorded at 100 mV/s in 0.5 M KOH. Each data point represents the average peak potential for desorption for more than 10 electrodes. The error bars indicate ± 1 standard deviation. The potentials are reported vs. Ag/AgCl/sat'd KCl.

lengths, the values of E_p were more positive for the partially formed layers than for the densely-packed layers. Comparable behavior was found for bilayers formed by varying the immersion time of thiol coated Au in solutions containing the surfactant CTAB. In all instances, E_p for the "bilayer" coated Au was more negative than that for the thiol coated Au. The value for E_p also progressed to more negative values for longer immersion times, a result which is consistant with the formation of a more densely packed bilayer structure. It is therefore clear that the shifts in E_p are not caused by a change in the reversible reduction potential.

The above dependence for the desorption of the layers can, however, be rationalized if the monolayer film is said to be resistant to ion transport to an extent that electron transfer to the thiol head group is limited by the rate at which ions can move through the film to the electrode surface. Such penetration is necessary to maintain charge neutrality as reduction proceeds. Under this condition, the onset of desorption is related to the potential required to establish a potential gradient through the hydrocarbon portion of the film that induces an ion flux sufficient to support the reduction reaction. If it is assumed that $E_{sol} - E_{appl}$ is dropped uniformly across the monolayer, the gradient necessary for desorption would be close to $(E_{sol} - E_p)/d$. Examination of Fig. 7 indicates that this is clearly not the case as it leads to the conclusion that as the thickness of the monolayer increases, the potential gradient required to sustain an ion flux sufficient for desorption decreases. If, for example, E_{sol} is said to equal pzc for a sulfur coated Au electrode ($\sim +0.25$ in vs. Ag/Ag/Cl(sat'd KCl) [18]), $(E_{sol} - E_p)/d$ is $\sim 2 \times 10^7$ V/cm for a propanethiol monolayer and 5×10^6 V/cm for a n-octadecanethiol film. These values indicate that the proposed model of a non-uniform potential drop through the n-alkanethiol film is more consistent with the data in Fig. 7. As shown below, if the potential drop within the hydrocarbon portion of the film is determined at E_p using eqns. (4) through (6), the potential gradient in the hydrocarbon phase at an electrode potential E_p is found to be independent of n.

In Fig. 8, values of ΔE_{MS} and ΔE_{HC} that were calculated from the data in Figs. 3 and 7 using eqns. (4) through (6) are plotted as a function of n. It can be seen that ΔE_{MS} decreases and ΔE_{HC} increases as n increases. The potential gradient within the hydrocarbon portion of the film at this potential is equal to $\Delta E_{HC}/d$. This quantity is plotted as a function of n in Fig. 9. Within the error of the experiment, $\Delta E_{HC}/d$ is essentially constant for monolayers with n greater than 7 and equals $\sim 3 \times 10^6$ V/cm. Presently, we do not understand these data for the short chain layers, but suspect that structural disorder may play a role. An upper limit of the ionic conductivity of the hydrocarbon phase, which is calculated by dividing the peak current density by the electric field, is 3×10^{-11} Ω^{-1} cm^{-1}. Because the surface coverage at E_p is less than the surface coverage of a well-formed monolayer, this value of the conductivity is actually an overestimation and more indicative of the conductivity of a partial monolayer. A change in the conductivity of the monolayer film as desorption occurs is manifested in the shapes of the voltammetric waves for the reductive desorption. For scan rates less than 250 mV/s, the peak width (typically ~ 45 mV) is narrower than that expected for a kinetically fast one

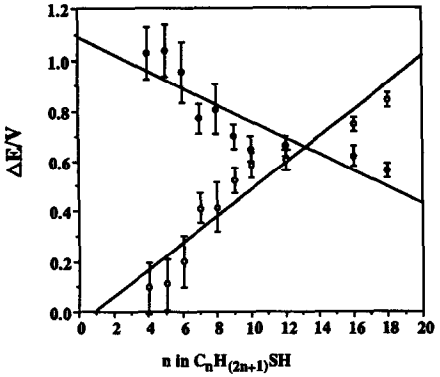


Fig. 8. The change in potential in the head-group (ΔE_{MS}, ○) and hydrocarbon tail-group (ΔE_{HC}, ●) at an n-alkanethiol derivatized Au/mica electrode for an applied electrode potential of E_p. The value of n corresponds to the number of carbons in the alkanethiol adsorbate, $C_nH_{(2n+1)}SH$. The points were determined from the data in Figs. 3 and 6 using eqns. (3) through (5). Error bars indicate ± 1 standard deviation.

electron surface process [38] which implies that, once initiated, further desorption becomes more favorable. Thus, although the overpotential required for desorption as n increases is adequately calculated from the change in E_p (as shown by the fact that the peak shape is not a strong function of n), the ion flux driven by the potential gradient $\Delta E_{HC}/d$ is more accurately given by the current density at the onset of desorption. Though difficult to estimate, the current density at the onset of desorption is likely to be comparable or less than the background charging current, i.e. 1 μA or less. This places the conductivity of the hydrocarbon portion of the layer in the range of 10^{-15} to 10^{-14} Ω^{-1} cm^{-1}. A more accurate assessment of the ionic conductivity of these layers may be found by performing controlled current electrolysis of the monolayer to examine the potential of the desorption as a

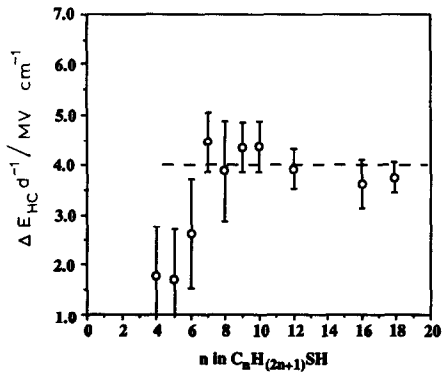


Fig. 9. The potential gradient within the hydrocarbon film as a function of n. The thicknesses of the hydrocarbon layers were calculated using eqn. (2). Error bars indicate ± 1 standard deviation.

function of the surface coverage. We note that the estimated conductivity is well within the range of those reported for bilayer lipid membranes, which have been found to vary from 10^{-10} to 10^{-16} Ω^{-1} cm^{-1} [39] and are are very sensitive to membrane composition.

The reductive desorption of n-alkanethiol monolayers on Au/mica was also examined in solutions containing LiOH or NaOH as the supporting electrolyte. Because the value of the peak potential is affected by uncompensated resistance, as evidenced by the linear relationship between i_p and E_p for voltammograms recorded at various scan rates, the intercept of the plot of E_p vs. i_p was taken as the "actual" peak potential, E_p^*, for the purpose of comparing the desorption potential in electrolytes of differing composition. This potential was measured in solutions of 0.5 M LiOH, NaOH, and KOH and is plotted vs. n in Fig. 10. In each case, the relationship between E_p^* and n is linear with the slopes of the best-fit lines for each data set increasing in the order KOH $\leqslant$ NaOH < LiOH. The relative size of the slopes indicates qualitatively that a greater overpotential is required for desorption in the presence of Li^+ relative to Na^+ or K^+. This may be due to stronger ion pairing of Li^+ in the organic phase or the existence of a larger barrier to Li^+ entering the film because it is more strongly solvated in solution. Alternatively, a steeper potential gradient may be necessary to induce a sufficient flux of Li^+ because it is less favorably partitioned within the hydrocarbon layer prior to desorption than is Na^+ or K^+. Within experimental error, the intercept of the best-fit line for each data set is equal to -0.85 V. This intercept can be considered as the potential change through the Au–S layer at the applied potential of E_p^* when n is equal to zero. The value of the intercept is not dependent on the ionic composition of the electrolyte because, to a first approximation, it is determined only by the properties of the Au–S bond.

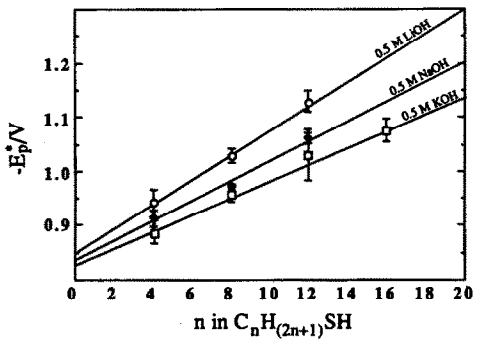


Fig. 10. The peak potential corrected for uncompensated resistance effects (E_p^*) vs. n for the reductive desorption of spontaneously adsorbed n-alkanethiol monolayers from Au/mica electrodes. The values of E_p^* are the intercepts of plots E_p vs. i_p from cyclic voltammograms recorded at various scan rates for the reductive desorption of a particular chain length n-alkanethiol monolayer. The value of n corresponds to the number of carbons in the n-alkanethiol adsorbate, $C_nH_{(2n+1)}SH$. Data are shown from voltammograms recorded in 0.5 M LiOH, 0.5 M NaOH, and 0.5 M KOH.

There are two assumptions in the above analysis that require qualification: C_{MS} being independent of both n and E_{app}. Qualitative justification of the latter assumption is found in the electrocapillary curves for Au surfaces coated with a monolayer of S [16]. In the potential range examined, it was found that for potentials more negative than the potential of zero charge (+0.2 V) and positive of the potential for desorption (−1.0 V), the capacitance (~ 10 to 15 $\mu F/cm^2$) of the electrical double layer was approximately independent of the electrode potential. In regard to the former assumption, one would not expect that the dielectric properties of the Au–S bond would be affected by the addition of more than the first few carbons in the hydrocarbon chain of the n-alkanethiol.

(III.E) Electrochemical characterization of n-alkanethiol monolayers at Au/Si and Au/glass surfaces

Monolayer films formed by the spontaneous adsorption of n-alkanethiols on Au/Si and Au/glass substrates were also examined briefly. Reductive and oxidative desorption of these layers was possible; however the voltammetric waves were ill-defined and not reproducible from sample to sample. The charge required for reduction of the adsorbed layer was typically 50% of that necessary for reduction of a monolayer at an Au/mica electrode. We do not believe that this reflects actual differences in the coverage of the n-alkanethiol adsorbate at the various surfaces, since, as discussed in Section III.B, the average structure and interfacial properties of n-alkanethiol films on Au/mica and Au/Si electrodes are comparable. It is more likely these differences are related to incomplete oxidation of the adsorbed n-alkanethiol or its oxidation to a Au–SR species that is not reduced within the available potential window. We suggest an inhomogeneity of the head-group chemistry for adsorption at Au/Si because we have found that the surface of these electrodes is more heterogeneous than the surface of a Au/mica film. As discussed in Section III.A, examination of the underpotential deposition and stripping of lead from Au/mica, Au/silicon, and Au/glass electrodes indicates the presence of chemically distinct sites on these surfaces. Au/mica films appear to have almost entirely Au(111) character, whereas Au/silicon and Au/glass electrodes contain Au(110) character as well. Though the average structure within the hydrocarbon portion of films on the various substrates appears to be unaffected, the reactivity of the thiol head-group may be more sensitive to substrate crystallinity.

(III.F) Spontaneously adsorbed n-alkanethiol layers at Ag/mica, Ag/Si, and Ag/glass electrodes

(III.F.1) Electrochemical desorption of n-alkanethiol monolayers from Ag electrodes

Voltammograms similar to those shown in Figs. 4B and 4C were also obtained for the reductive desorption of n-alkanethiols from Ag/mica, Ag/Si and Ag/glass electrodes. The potential and charge associated with the reduction, however, are different from that for the desorption from Au/mica samples. Only n-propane-,

n-butane-, n-pentane- and n-hexanethiol could be reduced within the potential window of Ag and reduction of the layer required 67 (± 15) $\mu C/cm^2$ regardless of the support material. The quantity of charge required for reduction is less than would be expected for the reduction of a complete monolayer commensurate with any of the low index planes [40] suggesting that monolayers formed at Ag are either less densely packed than would be predicted based on the assumption of an adsorbate film commensurate with the underlying Ag or that only a portion of the monolayer is electroactive (i.e. oxidized upon adsorption). The former possibility is contradicted by comparisons of the results of IRRAS, optical ellipsometry, and capacitance experiments examining monolayer films on Ag and those examining monolayers at Au; in all cases there is evidence for somewhat thicker ($\sim$ 10 to 20%) and higher density films at Ag [30,40] than at Au. We therefore believe that the value of σ_{rd} reflects inhomogeneity of the head-group binding, as has been discussed for monolayers at Au/Si and Au/glass electrodes. One possible explanation is that a portion of the layer is electroinactive at potentials accessible at Ag prior to solvent reduction. Another possibility is that a portion of the layer is not oxidized upon adsorption. The latter is suggested in consideration of the results of simultaneous electrochemical and radiotracer studies of sulfur adsorbed at single crystal silver surfaces [18]. In that study, silver electrodes were exposed to a mixture of $H_2{}^{35}S$ and H_2, which resulted in the formation of stable monomolecular films of sulfur. To examine the redox state of the adsorbed S, the quantity of radio-labelled S on the Ag surface was monitored as the electrode potential was scanned negatively. Between -0.5 and -0.6 V, loss of material from the Ag surface corresponded to 33% of the layer, but no Faradaic current was observed. Between -0.6 and -0.7 V, the remaining S was discharged with a concomitant cathodic current corresponding to 2 electrons per desorbed sulfur species. Adsorption of n-alkanethiols at Ag may be described similarly. If a portion of the layer is merely physisorbed (stabilized further by virtue of hydrophobic interactions) the charge required for desorption of the film would be less than 1 electron per adsorbed molecule.

As shown in Fig. 6, the peak potential for the reductive desorption at a Ag surface for a particular chain-length n-alkanethiol was 0.30 V negative of the desorption potential for the same n-alkanethiol at Au/mica. In other words, for the four compounds reduced within an accessible potential range at Ag, $\Delta E_p/n$ is equal to that for the desorption from Au/mica. This suggests that the electric field required to initiate ion motion through the hydrocarbon layer, to the extent that it can be assessed in this study, is approximately the same for monolayers at both Ag and Au surfaces implying that the partitioning and/or permeability to ions of the n-alkanethiol layer is not greatly affected by the substrate metal or the relatively small differences in the packing density. The intercept of the plot of E_p vs. n, however, is different for the two substrates.

The intercept of the plot of E_p vs. n is 0.30 V more negative for desorption from Ag compared to Au surfaces. We believe this is because of the differences in pzc for the two metals (-0.15 V for S–Ag at Ag(111) and $+0.25$ V for S–Au at Au(100) [18]), from which it follows that the potential required to develop the requisite

potential gradient for desorption at Ag is ~ 0.4 V more negative than at Au. The relatively large standard deviations for the data for the capacitance and charge for the reductive desorption for monolayers at Ag precludes a more detailed analysis of the similarities and differences in data in Fig. 7.

(III.F.2) Monolayer formation at Ag surfaces from solutions containing sodium n-octadecanethiolate

The electrochemical experiments outlined above indicate that for both Ag and Au substrates, formation of an n-alkanethiol monolayer requires the loss of the thiol hydrogen and oxidation of the sulfur upon adsorption for least a portion of the constituent n-alkanethiol molecules. The mechanism for adsorption at Ag surfaces, however, appears to be different from that for an Au surface. One indication of this is that the monolayers formed on Ag substrates from n-alkanethiol and n-alkanethiolate solutions, unlike films formed at Au from these adsorbate precursors, have similar structures. We believe that complete monolayers form at Ag from n-octadecanethiolate because the native surface of the Ag substrate is initially in an oxidized state as part of a silver oxide or silver halide film. Thus, the surface Ag(I) species can act as the oxidant to oxidize the thiol head-group. We envisage the adsorption process at Ag as:

$$AgX + RSH \rightarrow AgSR + X^- + H^+ \qquad (VI)$$

where X may be Cl, I, or OH. Although, as written, reaction (VI) describes a displacement reaction, consideration of the electronegativities of Cl, O, and S (3.5, 3.0 and 2.5, respectively [41]) suggest the reaction has electron transfer character. The relative electronegativities of these adsorbates indicate that there is less positive partial charge on the Ag in AgSR than in AgCl or AgOH; i.e. electron density is given to the Ag atom in reaction (VI). Preliminary results from Auger spectroscopic analyses confirm the above displacement mechanism.

(V) CONCLUSION

We have described the voltammetry associated with the electrochemical desorption of n-alkanethiol monolayers from gold and silver surfaces. Analysis of the voltammetry indicates that a substantial portion of the thiol groups in the monolayer are deprotonated upon adsorption and that the sulfur atom of the deprotonated thiol is oxidized by one electron. The presence of hydrogen peroxide in the aqueous solutions that were used for the adsorption of n-propanethiol onto high surface area gold samples supports this postulate. The agreement of the surface coverage determined from the charge required for reductive- and oxidative-desorption of n-alkanethiol monolayers from Au/mica electrodes with that predicted from the reported structure of n-octadecanethiol layers on Au(111) surfaces [18–21] suggests that this mechanism describes adsorption of all of the constituent n-alkanethiol molecules at Au/mica and the surface packing does not depend on the length of the hydrocarbon tail. Monolayers formed at Au/glass, Au/Si, and Ag

films at all support materials have been found to be only partially electroactive. We believe that this is related to bonding differences that arise from chemical heterogeneity of the surfaces of these metal films. To further investigate this possibility, infrared spectroelectrochemical studies examining n-alkanethiol films on polycrystalline and well-defined single crystal surfaces are planned.

Experiments in which monolayers were deposited on the various metal surfaces from deaerated solutions containing sodium n-octadecanethiolate in THF indicate that the presence of the hydrogen of the n-alkanethiol is required for the formation of a complete monolayer on Au. We postulate that this hydrogen furnishes an oxidant or takes part in the reduction reaction necessary for oxidation of the S species. On Ag substrates, where the surface is an oxide which can act as the oxidant, complete monolayers form when the substrate is exposed to the thiolate solution indicating that the thiol hydrogen is not necessary for adsorption at Ag.

Finally, we have been able to model the behavior of the electric double layer at these derivatized electrodes by examining the chain-length dependence of both the capacitance and the peak potential for reductive desorption. The results indicate that the surface film acts as would two capacitors that are connected in series where one capacitor corresponds to the thiol head-group region of the monolayer and the other to the hydrocarbon phase. A potential gradient of approximately 3.5×10^5 V/cm in the hydrocarbon region is required to allow sufficient ion flux through the hydrocarbon layer for electro-desorption to proceed at both Au and Ag electrodes indicating that the ionic conductivity of this portion of the layer is in the range of 10^{-13} to 10^{-15} Ω^{-1} cm^{-1}. This is within the range of conductivities reported for bilayer lipid membranes. These findings are relevant to a number of studies in which electron transfer processes at monolayer coated electrodes have been examined [7,42–48].

ACKNOWLEDGEMENTS

M.D.P. gratefully acknowledges support of a Dow Corning Assistant Professorship. We thank Dr. A.J. Bevolo for the Auger measurements. Ames Laboratory is operated for the U.S. Department of Energy by Iowa State University under Contract No. W-7405-eng-82. This work is supported by the Office of Basic Energy Sciences, Chemical Science Division.

REFERENCES

1 J.O. Swalen, D.L. Allara, J.B. Andrade, E.A. Chandross, S. Garoff, J. Israelachvili, T.J. McCarthy, R. Murray, R.F. Pease, J.F. Rabolt, K.J. Wynne and H. Yu, Langmuir, 3 (1987) 932.
2 J. Sagiv, Isr. J. Chem., 78 (1979) 339, 346.
3 J. Sagiv, J. Am. Chem. Soc., 102 (1980) 92.
4 S.R. Wasserman, T.Y. Tao and G.M. Whitesides, Langmuir, 5 (1989) 1074.
5 D.L. Allara and R.G. Nuzzo, Langmuir, 1 (1985) 45, 52.
6 R.G. Nuzzo and O.L. Allara, J. Am. Chem. Soc., 105 (1983) 4481.
7 M.D. Porter, T.B. Bright, D.L. Allara and C.E.D. Chidsey, J. Am. Chem. Soc., 109 (1987) 3559.

8 C.D. Bain, E.B. Troughton, Y.T. Tao, J. Evall, G.M. Whitesides and R.G. Nuzzo, J. Am. Chem. Soc., 111 (1989) 321.
9 R.G. Nuzzo, L.H. Dubois and D.L. Allara, J. Am. Chem. Soc., 112 (1990) 558.
10 G.M. Whitesides and F.E. Laibinis, Langmuir, 6 (1990) 87.
11 L.C.F. Blackman, M.J.S. Dewar and H. Hampson, J. Appl. Chem., 7 (1957) 160.
12 C.J. Sandaroff, S. Garoff and K.P. Lenny, Chem. Phys. Lett., 96 (1983) 547.
13 T.H. Hoo, K. Kim and M.S. Kim, J. Phys. Chem., 90 (1986) 5816.
14 R.L. Sobocinski, M.A. Bryant and J.E. Pemberton, J. Am. Chem. Soc., 113 (1991) 3629.
15 R.G. Nuzzo, B.R. Zegarski and L.H. Dubois, J. Am. Chem. Soc., 109 (1987) 733.
16 D.G. Wierse, M.M. Lohrengel and J.W. Schultze, J. Electroanal. Chem., 92 (1978) 1215.
17 E. Lamy-Pitara and J. Barbier, Electrochim. Acta, 31 (1986) 717.
18 C. Nguyen Van Huong, R. Parsons, P. Marcus, S. Montes and J. Oudar, J. Electroanal. Chem., 119 (1981) 137.
19 C.E.D. Chidsey, G.-Y. Liu, P. Rowntree and G. Scoles, J. Chem. Phys., 91 (1989) 4421.
20 L. Strong and G.M. Whitesides, Langmuir, 4 (1988) 546.
21 C.A. Widrig, C.A. Alves and M.D. Porter, J. Am. Chem. Soc., 113 (1991) 2805.
22 D.D. Perrin, W.L.F. Armarego and D.R. Perrin, Purification of Laboratory Chemicals, 2nd ed., Permagon Press, Ithaca, NY, 1980, p. 426.
23 M.S. Zei, Y. Nakai, G. Lehmpfuhl and D.M. Kolb, J. Electroanal. Chem., 150 (1983) 201.
24 T.M. Gardiner and M.H.B. Stiddard, Thin Solid Films, 77 (1981) 335.
25 R.J. Tucceri and D. Posadas, J. Electroanal. Chem., 270 (1989) 415.
26 H. Jaeger, P.D. Mercer and R.G. Sherwood, Surf. Sci., 11 (1968) 265.
27 C.E.D. Chidsey, D.N. Liocano, T. Sleator and S. Nakahara, Surf. Sci., 200 (1988) 45.
28 V.N. Hallmark, S. Chiang. J.F. Rabolt, J.D. Swalen and R.J. Wilson, Phys. Rev. Lett., 59 (1987) 2879.
29 J.W. Schultze and D. Dickertman, Surf. Sci., 54 (1976) 489 and (b) K. Engelsman, W.J. Lorenz and E. Schmidt, J. Electroanal. Chem., 114 (1980) 1.
30 A. Bewick and B. Thomas, J. Electroanal. Chem., 65 (1975) 91.
31 M.N. Walczak, C.K. Chung, S.M. Stole, C.A. Widrig and M.D. Porter, J. Am. Chem. Soc., 113 (1991) 2370.
32 R.G. Snyder, S.L. Hsu and S. Krimm, Spectrochim. Acta, Part A, 34 (1978) 395; R.G. Snyder, H.L. Strauss and C. Ellinger, J. Phys. Chem., 86 (1982) 5145.
33 R.G. Greenler, J. Phys. Chem., 44 (1966) 310.
34 Ionization Constants of Organic Compounds in Aqueous Solutions, IUPAC Chemical Data Series No. 23, Permagon Press, Oxford, 1983.
35 F.W. Sears, M.W. Zemansky and H.O. Young, University Physics, Addison–Wesley, Reading, MA, 1984, p. 519.
36 A.J. Bard, L.R. Faulkner, Electrochemical Methods: Fundamentals and Applications, Wiley, New York, 1980, p. 507.
37 E.E. Polymeropoulus and J. Sagiv, J. Chem. Phys., 69 (1978) 1836.
38 Ref. 36, pp. 409–413.
39 H.T. Tien, Bilayer Lipid Membranes (BLM): Theory and Practice, Marcel Dekker, New York, 1974, pp. 118, 511–512.
40 M.N. Walczak, C. Chung, R.S. Deinhammer and M.D. Porter, in preparation.
41 C.E. Mortimer, Chemistry: A Conceptual Approach, 4th ed., Van Nostrand, New York, 1979.
42 H.O. Finklea, S. Avery, M. Lynch and T. Furtsch, Langmuir, 3 (1987) 409.
43 E. Sabatani and I. Rubinstein, J. Phys. Chem. 91 (1987) 6663.
44 I. Rubinstein, S. Steinberg, Y. For, A. Shanzer and I. Sagiv, Nature, 332 (1988) 426.
45 H.O. Finklea, O.R. Snyder and J. Fedyk, Langmuir, 6 (1990) 371.
46 C.E.O. Chidsey and D.N. Loiacono, Langmuir, 6 (1990) 682.
47 C.E.O. Chidsey, C.R. Bertozzi, T.M. Putvinski and A.M. Mujsce, J. Am. Chem. Soc., 112 (1990) 4301.
48 S.E. Creager, G.K. Rowe, Anal. Chim. Acta, 246 (1991) 233.